

Machine Learning Model for the Prediction of Slow Vital Capacity

Samad Jahandideh¹, Albert A. Taylor¹, Amy Bian², Lisa Meng²,
Danielle Beaulieu¹, Mike Keymer¹, Jinsy Andrews², David L. Ennist¹



¹Origent Data Sciences, Inc., Vienna, VA, USA
²Cytokinetics, Inc., South San Francisco, CA, USA



Introduction

Background: This report is the first step of a research collaboration that aims to retrospectively validate an ALS disease progression computer model leveraging data from a clinical trial of the investigational drug, *tirasemtiv* (BENEFIT-ALS). The PRO-ACT database is a rich source of Forced Vital Capacity (FVC) records, but sparse Slow Vital Capacity (SVC) records. BENEFIT-ALS used SVC as an outcome, thus we asked whether a reliable SVC predictive model could be developed using a training dataset rich in FVC records.

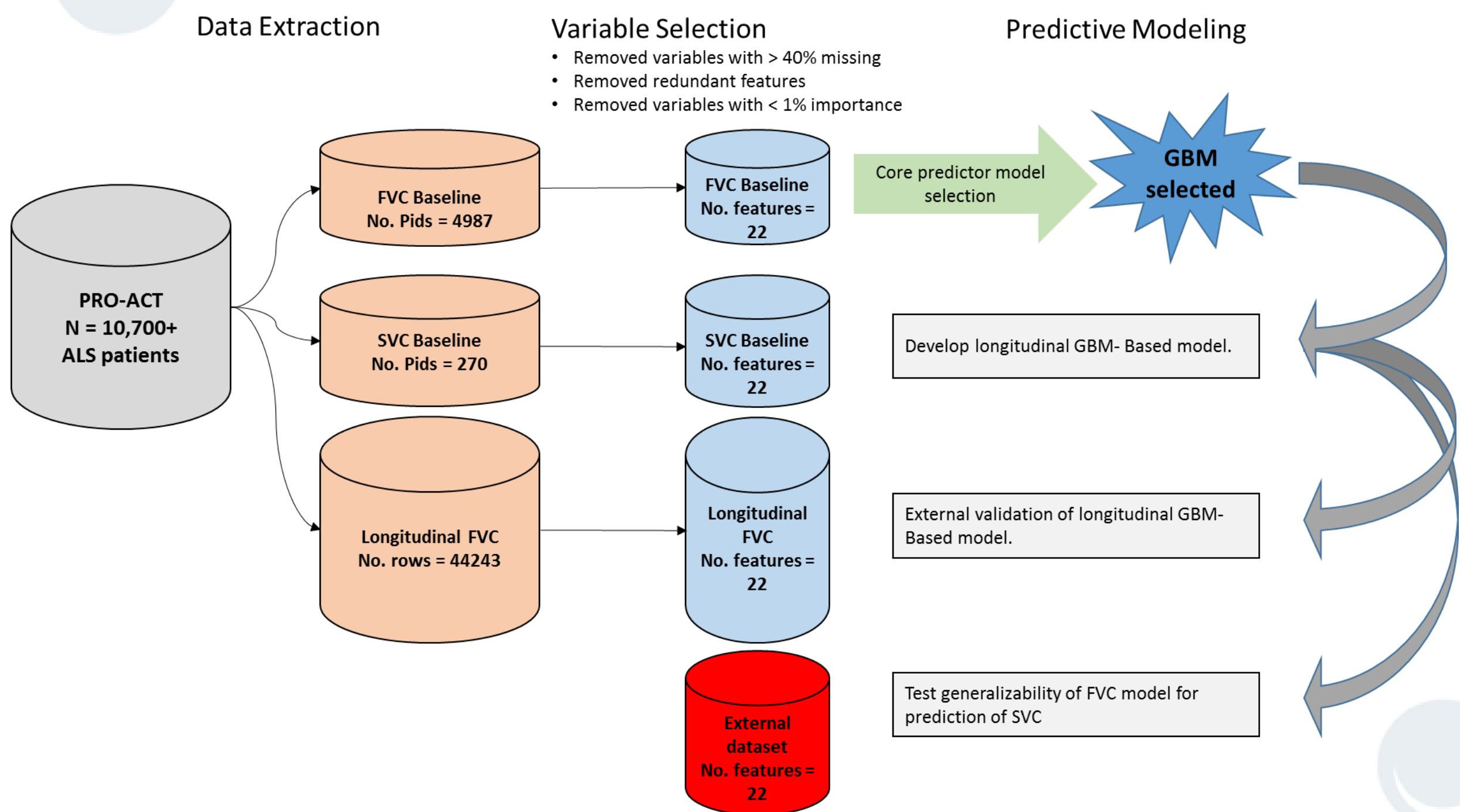
Objectives: To develop a model that predicts SVC using the PRO-ACT database.

Methods

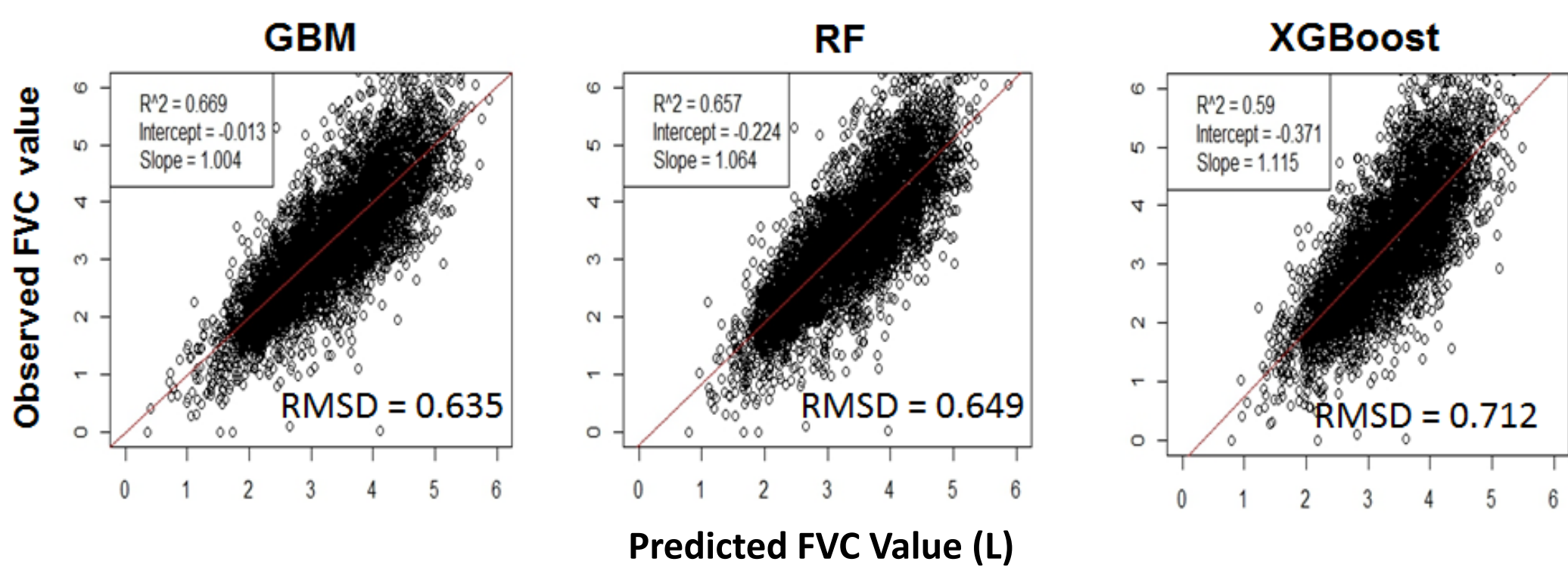
The PRO-ACT ALS database was used to compare the performance of three different predictive models, i.e. Random Forest¹ (RF), Gradient Boosting Machine² (GBM) and XGBoost³. The XGBoost option was discarded because it exhibited both a higher root mean square deviation (RMSD) and lower R² value than the other algorithms. Upon additional testing, it was revealed that the GBM model slightly outperformed the RF model.

Using the GBM algorithm, we compared predictions using a 30-day run-in period to predictions made from baseline without a run-in period, examined the useful range of the model (in terms of vital capacity and over time) and validated the GBM models with an external data set. Finally, we examined the generalizability of using the model for predicting SVC.

Workflow of VC Prediction



Performance Comparison of Machine-Learning Models

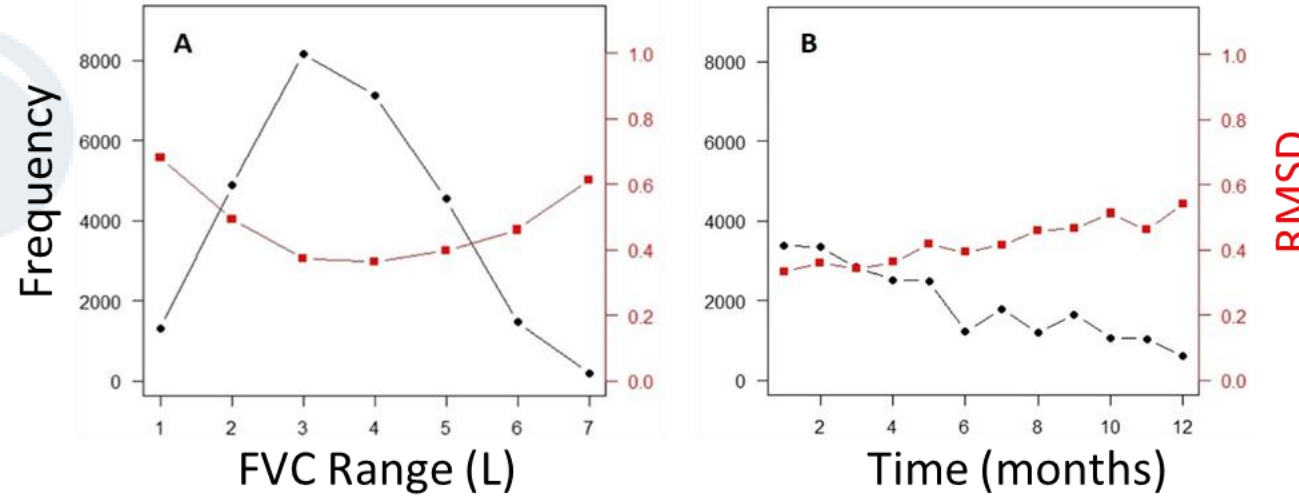


- Models were developed using data from baseline (N = 4987).
- 10-fold cross validation was applied to training and testing of models.

Conclusion:

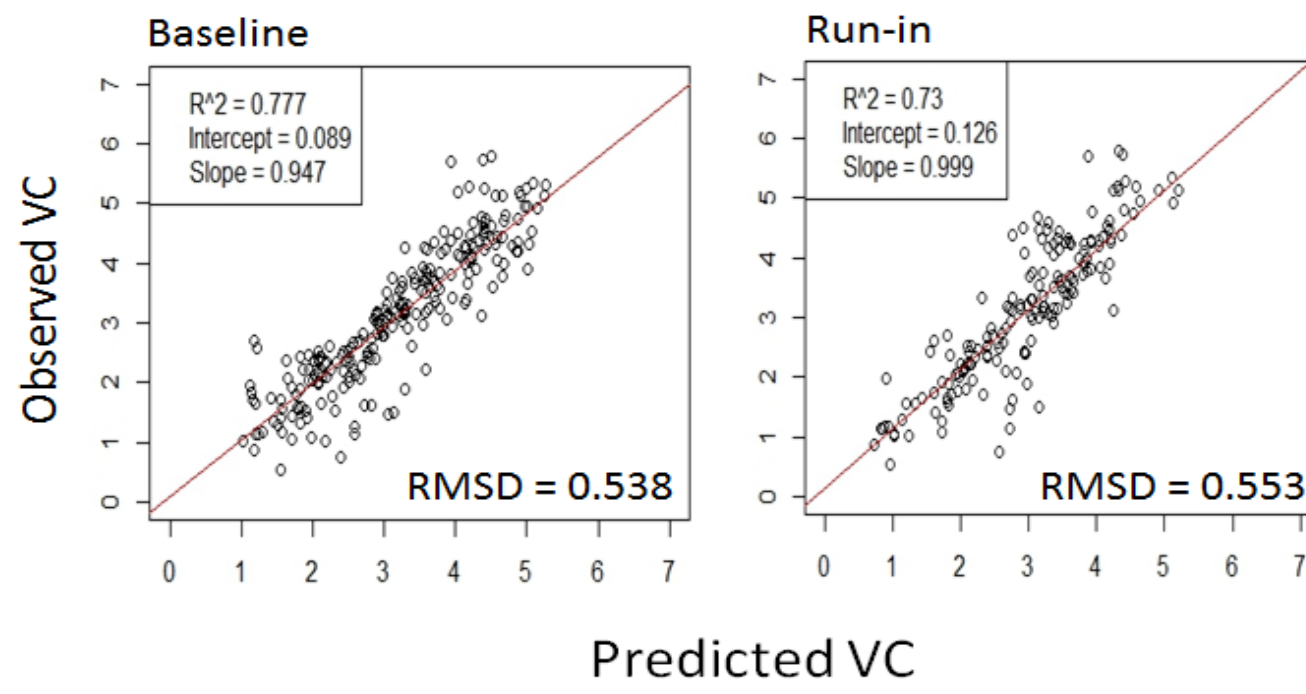
- The XGBoost option was discarded because it exhibited both a higher RMSD and lower R² value than the other algorithms.
- The GBM model slightly outperformed the RF model.

External RMSD as a Function of FVC Range and Time Interval



- RMSD increases for extreme FVC values
- RMSD increases with time
- Higher RMSDs are correlated with sparser data

External Validation

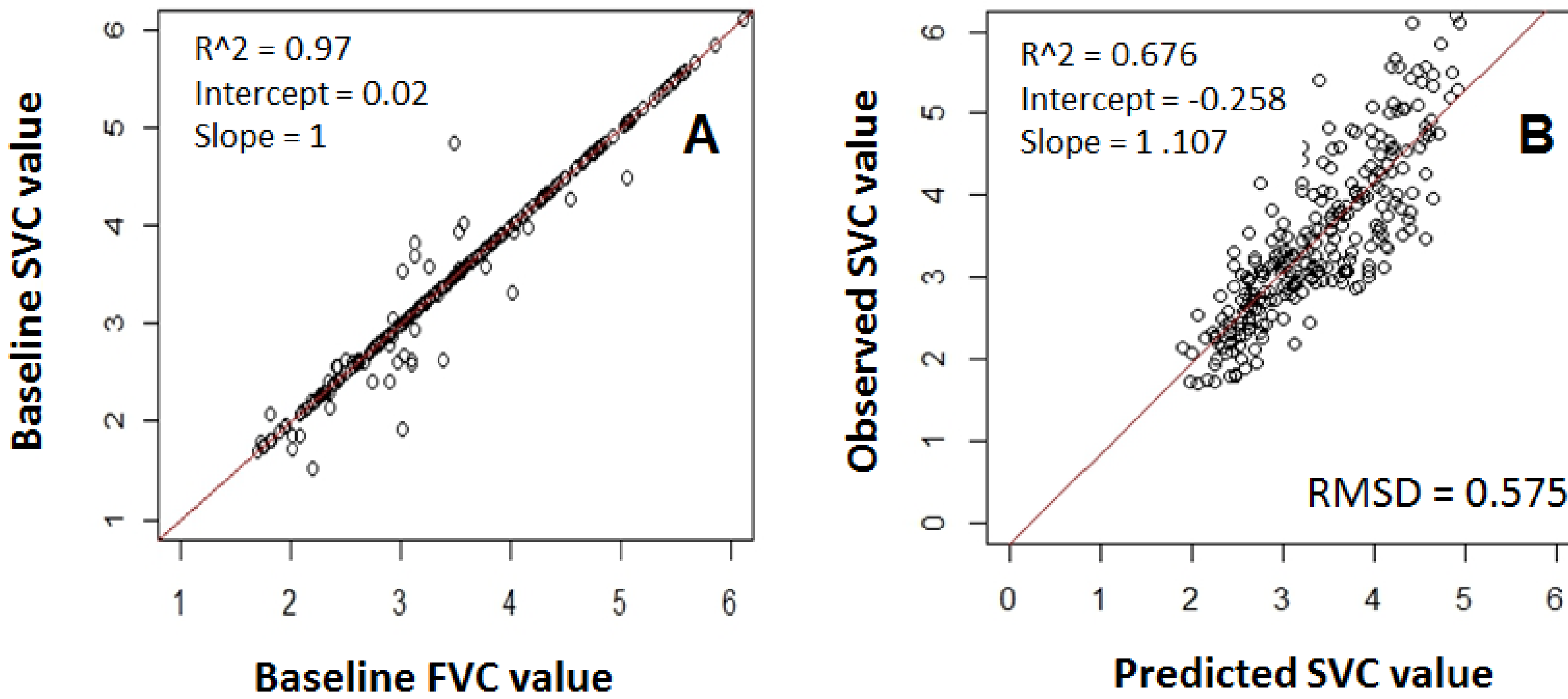


External dataset included 60 participants from trial NCT01257581 (courtesy of Dr. Nazem Atassi, MGH, see details on ClinicalTrials.gov; treated with creatine or tamoxifen).

Results

We developed a generalizable GBM-based predictive model for prediction of Vital Capacity. The longitudinal predictive model was validated using an external validation dataset.

Generalizability of FVC Model to Prediction of SVC



- A: Correlation between FVC and SVC values in 270 patients with both FVC and SVC records.
- B: Application of FVC model to prediction of SVC records.

Conclusions

We hypothesized the possibility of using FVC records to predict SVC scores of ALS patients using machine learning techniques. Our results support our hypothesis and showed acceptable correlation. GBM outperformed other models and we selected GBM as our core model for developing a tool to predict SVC.

References

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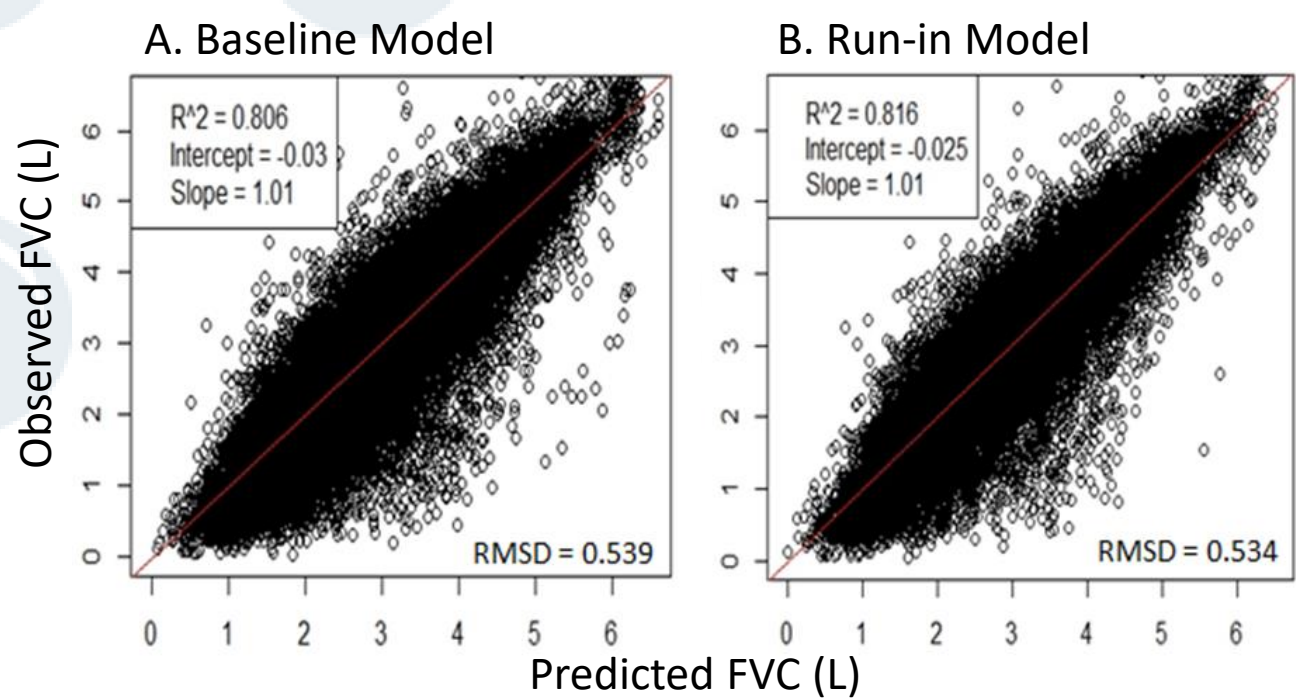
Acknowledgements

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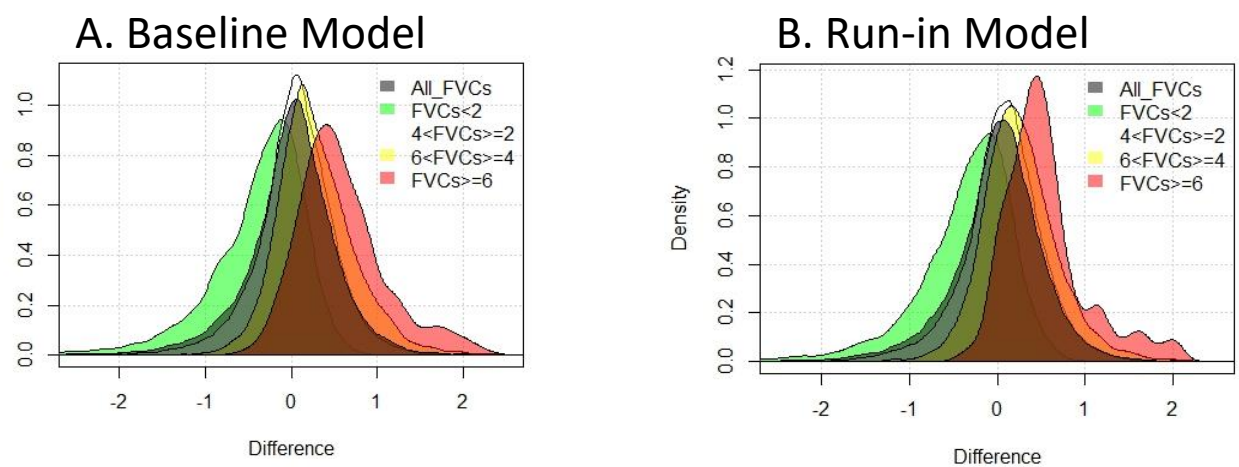
Contact

David L. Ennist, PhD, Chief Science Officer, dennist@origent.com

Comparison of Baseline and Run-in GBM Models



Difference Between Predicted and Observed FVC Values



Relative Importance of Predictor Variables in GBM Models

A. Baseline Model		B. Run-in Model	
Variable	Relative Importance (%)	Variable	Relative Importance (%)
FVC baseline	76.59	FVC baseline	77.11
Time since baseline	7.9	Time since baseline	6.89
Height baseline	2.73	Height baseline	2.59
ALSFRRS-R slope baseline	1.63	Speech subscore baseline	1.65
Time since onset	1.33	Time since onset	1.48
Gross subscore baseline	1.31	Gross subscore baseline	1.43
Speech subscore baseline	1.24	ALSFRRS-R slope baseline	1.42
Gender	1.24	Gender	1.25
Bulbar subscore baseline	1.14	Age	0.96
Age	0.98	Bulbar subscore baseline	0.88
Creatinine baseline	0.66	Creatinine baseline	0.79
ALSFRRS-R total baseline	0.55	Pulse baseline	0.69
Phosphorus baseline	0.54	ALSFRRS-R total baseline	0.65
Pulse baseline	0.51	Weight baseline	0.61
Weight baseline	0.49	Phosphorus baseline	0.53
BMI baseline	0.32	Time since diagnosis	0.32
Diastolic BP baseline	0.29	BMI baseline	0.28
Time since diagnosis	0.27	Diastolic BP baseline	0.25
Respiratory Insufficiency score baseline	0.13	Systolic BP baseline	0.12
Systolic BP baseline	0.09	Chest subscore baseline	0.06
Chest subscore baseline	0.06	Respiratory Insufficiency score baseline	0.04

